

An Investigation of Foot Temperature Deviations in Individuals with Diabetes: Insights from Wearable In-Shoe Technology*

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Abstract—Plantar foot temperature is a valuable indicator of diabetes-related complications, but traditional assessment methods, such as infrared thermography and contact thermometers, require unshod feet and controlled conditions, limiting their practicality for continuous monitoring. In this study, we employ a smart insole with 21 embedded temperature sensors to capture plantar temperature data from shod feet. We introduce a novel approach that leverages per-foot relative temperature values—normalized to the foot’s mean—rather than absolute values or inter-foot asymmetry. Using data collected during static postures (lying, sitting, and standing), we evaluate multiple machine learning classifiers, with Random Forest achieving the highest accuracy (83.20%), alongside high sensitivity (93.75%) but moderate specificity (63.6%). To enhance explainability, we apply SHAP analysis to interpret model predictions and identify key sensor contributions. Additionally, we derive simple decision rules from the Random Forest model, finding that two medial arch sensors can achieve near-equivalent accuracy (80.38% and 79.82%) to the full model. These results suggest that deviations in plantar temperature patterns could serve as an indicator of diabetes-related foot health changes. Future work will expand this approach to ambulatory activities, integrating static and dynamic features to develop an insole-based system for continuous foot health monitoring in real-world settings.

Clinical relevance— This study demonstrates that plantar temperature patterns, captured via smart insoles, can reflect changes associated with diabetes-related foot health conditions. The identification of simple, high-accuracy decision rules using medial arch sensors suggests a potential low-cost, non-invasive tool for detecting thermal irregularities linked to diabetes-related foot complications.

I. INTRODUCTION

Plantar foot temperature can serve as a meaningful early indicator of the onset of certain medical conditions caused by diabetes mellitus (DM), such as foot ulceration and peripheral artery disease (PAD) [1], [2], [3].

Infrared thermography is a well-established method for assessing plantar temperature [2], [4], [5], [6], [7]. However,

this approach has notable limitations. It requires subjects to be unshod, remain stationary, and undergo assessment in a controlled clinical setting, making it impractical for continuous monitoring in daily life.

Wearable technologies, particularly smart insoles, offer a promising alternative [3], [8], [9]. These devices enable continuous temperature monitoring while users remain shod and engaged in daily activities, providing a more accurate reflection of real-world conditions.

A. Related Work

In machine learning (ML)-based research, a substantial body of work has utilised thermograms of unshod feet for diabetes prediction or foot ulcer detection [7], [10], [11], [12].

For wearables and shod feet, several studies have explored diabetes prediction using insoles equipped with pressure sensors [13], [14].

However, despite strong evidence suggesting that temperature-related features in shod feet can also help predict diabetes [3], ML-based studies on temperature-sensing insoles remain limited. To the best of our knowledge, only one study has explored the application of ML to foot temperature data collected through insoles [15].

Khandakar et al. [15] employed an insole with 8 temperature sensors to collect data from 43 participants in a standing position. The resulting temperature map was then fed into an artificial neural network for classification, distinguishing between diabetic and healthy individuals. Their study focuses exclusively on diabetic patients with severe cases of diabetic sensorimotor polyneuropathy (DSPN), uses a limited number of insole sensors, and lacks a comparison with other ML approaches.

B. Objectives

We employ a smart insole equipped with a dense array of temperature sensors, enabling the generation of detailed heat maps that can capture localized temperature variations across the plantar surface.

We focus on per-foot relative temperature values, i.e., we consider temperature values with respect to the mean temperature of the foot, unlike other works that either use absolute temperature values or look for asymmetry in temperature values across the feet. We determine that this approach can provide new insights on the differences between healthy and diabetic thermal patterns.

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We apply and evaluate a range of ML algorithms to identify the best-performing classifier for diabetes prediction. In this preliminary study, we use continuous temperature data collected during static postures (lying, sitting, and standing) to train the ML classifiers.

II. METHODS

A. Experimental setup

The data used in this study was obtained from a controlled observational study involving 101 participants, 74 of which suffered from diabetes, and 27 healthy controls with no history of significant medical, surgical, vascular, or neurological conditions.

Peripheral arterial perfusion, a key factor in assessing peripheral artery disease (PAD), was evaluated by an experienced clinician (with over 10 years experience in the field) using Doppler waveform analysis, ankle-brachial pressure index (ABPI), and toe-brachial pressure index (TBPI). Eligible participants exhibited triphasic waveforms, ABPI values between 1 and 1.29 and TBPI > 0.7 . Individuals with a history of smoking or alcohol abuse were excluded from the study.

The wearable system, developed by Tarsos [16], used in this study consisted of a temperature array in the form of an insole, with 21 embedded thermistors, arranged as shown in Fig. 1. These sensors transmitted data wirelessly via Bluetooth to a mobile app, with a sampling rate of 0.1 Hz. The thermistors respond rapidly to temperature changes, and the chosen sampling rate allows ample time for the sensors to reach thermal equilibrium with the foot surface. Different insole sizes were provided to suit different foot sizes.

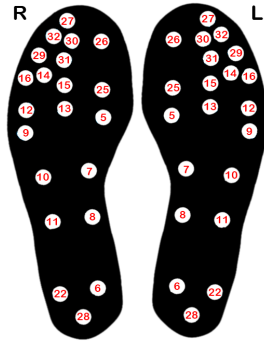


Fig. 1: Position and numbering scheme of the 21 temperature sensors on each insole

During data collection, participants wore shoes with the smart insoles and lay in a supine position for 10 minutes. This was followed by 10 minutes in a seated position, then 10 minutes standing. Prior to data capture, a 15-minute acclimatization period was allocated.

This research was approved by the University of Malta Research Ethics Committee. Written informed consent was obtained from all the participants. All research efforts conducted on humans were in compliance with the World Medical Association's Declaration of Helsinki.

III. RELATIVE TEMPERATURE DIFFERENCES

Despite the allotted time for foot acclimatization and all participants reaching thermal equilibrium with shod feet, some natural variation in foot temperatures still remained at the start of the recording sessions, with some individuals exhibiting warmer feet than others.

To minimize this variation and reduce potential bias from the participants' initial foot temperatures, we focus on relative temperature differences. Specifically, for each participant, the individual sensor values are adjusted by subtracting them from the mean temperature of that participant's foot. This approach allows for a more consistent comparison across participants.

Fig. 2 shows an example for one of the participants, where Fig. 2(a) shows the absolute temperature values measured by the insole, and Fig. 2(b) shows the relative temperature values relative to the mean foot temperature. The mean foot temperature is computed from all the 21 insole sensors on each foot, computed separately for each participant and foot.

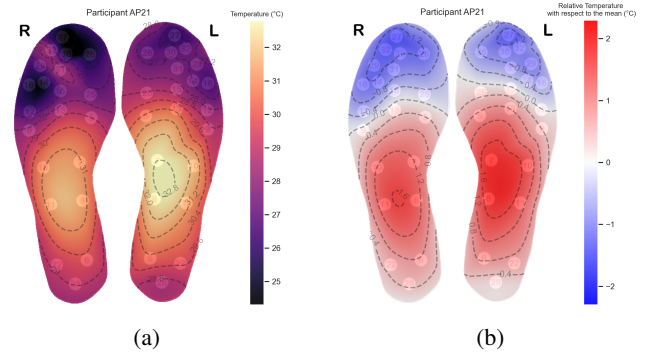


Fig. 2: (a) Absolute temperature values, and (b) Relative temperature values with respect to the mean foot temperature.

IV. DATA FEATURES

In various studies [3], [4], [7], [10], [11], [17], numerous features have been derived from temperature data, with different approaches explored for feature extraction.

For this study, we focus on using the temperature values directly as features, specifically relative temperature differences, as our primary interest is in the static poses where temperature remains relatively constant over time. This investigation is of particular relevance because for most individuals a significant portion of the day is typically spent in one of these three static poses (lying down, sitting, standing). While in future work we aim to expand and build upon dynamic analysis of ambulatory activity [18], in this work we focus specifically on the static aspects of foot temperature patterns.

V. CLASSIFICATION MODELS

We frame the machine learning task as a binary classification problem: distinguishing healthy individuals from participants with diabetes based on the foot temperature data captured using insoles.

We evaluated a diverse set of classifiers: Random Forest, Extra Trees, Histogram-based Gradient Boosting, Support Vector Machine (SVM), Quadratic Discriminant Analysis (QDA), and Logistic Regression. These classifiers were selected for their ability to handle high feature correlation and overlapping class distributions while providing effective decision boundaries.

The input feature set consisted of the vector:

$$\mathbf{x} = (s_1^L, s_2^L, \dots, s_{21}^L, s_1^R, s_2^R, \dots, s_{21}^R) \quad (1)$$

where s_i^L represents the relative temperature value of the i -th sensor on the left foot, and s_j^R represents the corresponding value for the j -th sensor on the right foot. This concatenation of left foot and right foot sensor values results in a feature vector $\mathbf{x}$ of length 42.

Temperature data from all recorded static poses (lying down, sitting, and standing) were used as input to the classifiers. The resulting feature matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$, comprises all feature vectors $\mathbf{x}$, where $N = 30,508$ represents the number of samples (rows), and $d = 42$ denotes the number of features. The classifiers were implemented using the Scikit-learn [19] and SciPy [20] Python libraries.

VI. MODEL TRAINING & PERFORMANCE EVALUATION

A. Addressing Class Imbalance

Since the dataset is imbalanced (77.5% diabetic, 22.5% healthy), class weights were adjusted during model training to mitigate this imbalance. For Random Forest and Extra Trees, instead of using class weights derived from the entire dataset, weights were computed separately for each bootstrap sample (subsample) used to construct individual trees. This ensures the weights reflect the actual distribution within each training sample, leading to a more balanced learning process.

B. Performance Evaluation

Nested stratified K -fold cross-validation was employed to evaluate the performance of the classifiers – see Fig. 3. The inner cross-validation loop handled model selection (hyperparameter tuning), while the outer cross-validation loop measured the model's performance – in this way an *unbiased estimate* of a model's performance is obtained [21], [22]. The number of folds was set to $K = 10$ for both loops.

Since the dataset consists of time-series data, with multiple samples per participant recorded at different time points, it was ensured that when the dataset is split into folds, all samples from a participant remain within the same fold. This prevents data leakage, which would occur if the folds overlapped, given that the samples for a given participant are highly correlated.

C. Hyperparameter Tuning

For hyperparameter tuning, we performed a *grid search* over the search space outlined in Table I, with the optimal values also provided in the same table.

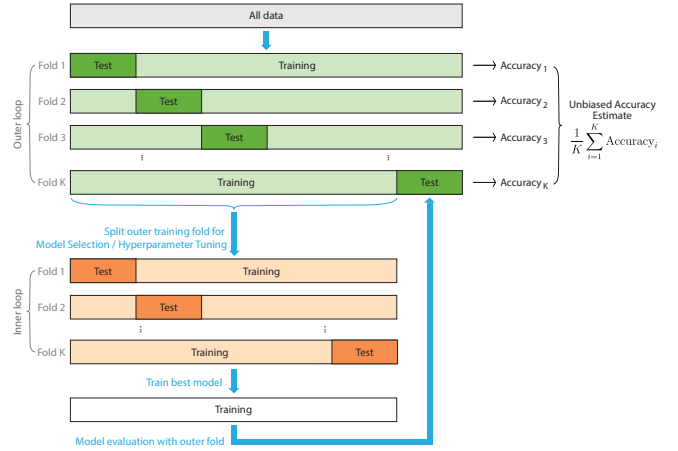


Fig. 3: Nested K -fold Cross Validation

D. Model Stability

After identifying the best classifier in the previous step, namely, the one with the highest unbiased accuracy, and after determining its optimal hyperparameters, we assessed the model's stability and consistency by performing 25 trials, each consisting of a 10-fold cross-validation. The performance metrics from all trials were averaged, and the standard deviation was calculated to evaluate the model's stability across different data splits. Additionally, confusion matrices were accumulated over all trials to provide insight into the model's classification behavior. Finally, the model selected for subsequent analysis was the one with accuracy closest to the mean accuracy across all trained models from the repeated trials.

VII. RESULTS

Table II presents the unbiased accuracy results for the different classifiers. The Random Forest classifier outperformed the others, achieving a mean accuracy of 83.20%. The standard deviation of 7.85% reflects the variation in accuracy across the test folds of the outer loop of the nested K -fold cross-validation.

After tuning the hyperparameters of the Random Forest model ($N_t = 140$, $d = 6$, cf. Table I) and conducting 25 trials with the optimized model to assess its stability as outlined in Section VI-D, we observed a reduction in the standard deviation of the model's performance, which decreased to 1.11%. This reduction highlights the model's stability and robustness to variations in data partitioning.

Since the Random Forest model was the best performing classifier, the remainder of this section will focus on its detailed evaluation.

Fig. 4(a) shows the normalized confusion matrix of the Random Forest model accumulated over the 25 trials, where TPR = true positive rate, FPR = false positive rate, FNR = false negative rate, and TNR = true negative rate.

The model exhibits a high sensitivity ($TPR = \frac{TP}{TP+FN}$) of 93.75% and a moderate specificity ($TNR = \frac{TN}{TN+FP}$) of 63.36%. This means the model is effective at detecting

TABLE I: HYPERPARAMETER TUNING

CLASSIFIER	HYPERPARAMETER SEARCH SPACE	OPTIMAL VALUES
Random Forest	number of trees $N_t \in \{40 + 10k \mid k = 0, 1, 2, \dots, 13\}$ max tree depth $d \in \{2 \dots 6\}$	$N_t = 140$ $d = 6$
Extra Trees	number of trees $N_t \in \{40 + 10k \mid k = 0, 1, 2, \dots, 13\}$ max tree depth $d \in \{2 \dots 6\}$	$N_t = 90$ $d = 6$
Histogram-based Gradient Boosting	learning rate $\eta \in \{0.001, 0.01, 0.1, 0.2\}$ max tree depth $d \in \{2 \dots 6\}$	$\eta = 0.1$ $d = 3$
Support Vector Machine (SVM)	kernel $K \in \{\text{linear, sigmoid, poly, rbf}\}$ regularisation parameter $C \in \{10^{-2}, 10^{-1}, 10^0, 10^1, 10^2\}$ polynomial kernel degree $p \in \{2 \dots 5\}$ polynomial / sigmoid independent term $c_0 \in \{0, 0.1, 0.5, 1, 10\}$ gamma $\gamma \in \left\{ \frac{1}{n}, \frac{1}{n \times \text{Var}(\mathbf{X})} \right\}$ where $\mathbf{X} \in \mathbb{R}^{m \times n}$	$K = \text{rbf}$ $C = 10$ – – $\gamma = \frac{1}{n}$
Quadratic Discriminant Analysis (QDA)	regularisation parameter $C \in \{0.01, 0.1, 0.0, 0.5, 1\}$	$C = 0.1$
Logistic Regression	regularisation parameter $C \in \{10^{-2}, 10^{-1}, 10^0, 10^1, 10^2\}$	$C = 0.01$

TABLE II: MODEL EVALUATION RESULTS

CLASSIFIER	TRAIN ACCURACY (MEAN $\pm$ S.D.)	TEST ACCURACY (MEAN $\pm$ S.D.)
Random Forest	97.90% $\pm$ 0.65%	83.20% $\pm$ 7.85%
Extra Trees	98.85% $\pm$ 2.40%	80.93% $\pm$ 9.42%
Histogram-based Gradient Boosting	99.96% $\pm$ 0.04%	79.09% $\pm$ 8.48%
Support Vector Machine (SVM)	98.97% $\pm$ 0.53%	81.51% $\pm$ 10.70%
Quadratic Discriminant Analysis (QDA)	99.64% $\pm$ 0.07%	78.49% $\pm$ 12.59%
Logistic Regression	97.87% $\pm$ 0.68%	77.08% $\pm$ 14.68%

foot deviations in diabetic patients, which is beneficial for the early identification of potential complications like PAD or foot ulcers. However, the moderate specificity means that 37% of non-diabetic individuals may be misclassified as suffering from diabetes-related complications, which can lead to some anxiety for the end users and the need (and costs) of further testing.

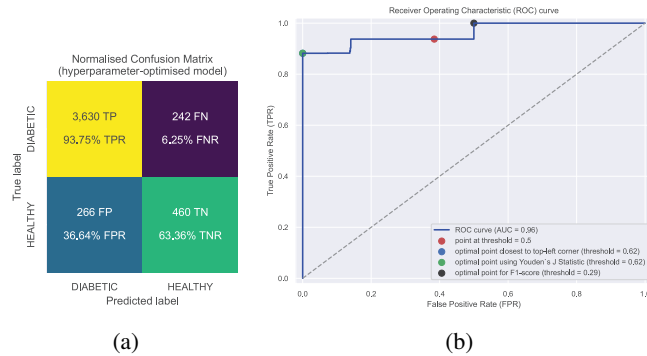


Fig. 4: (a) Confusion Matrix, and (b) ROC curve of the hyperparameter-optimized Random Forest model.

Seen from a clinical perspective, when used as pre-screening tool, prioritizing high sensitivity can be beneficial, as detecting diabetes-related foot health issues early may be more critical than reducing false positives. One has also to keep in mind that the model has been trained on static

poses only: we envision that by extending this work to cover ambulatory activity, together with the use of dynamic features and continuous monitoring over longer durations, all this may help to improve the overall accuracy of the model. Integrating other biomarkers or patient history alongside foot temperatures may also help to improve model specificity.

Fig. 4(b) shows the *receiver operating characteristic* (ROC) curve for the Random Forest model, illustrating how TPR varies with respect to FPR. The *area under the curve* (AUC) is 0.962.

The red point on the ROC curve corresponds to a threshold of 0.5 and reflects the FPR and FNR values shown in the confusion matrix in Fig. 4(a). The green point marks the optimal threshold identified by Youden's J statistic [23]. This maximizes the difference between TPR and FPR, defined as $J = \arg \max (TPR - FPR)$. This point also happens to coincide with the point on the ROC curve closest to (0,1), the perfect classifier. The black point on the ROC curve corresponds to the threshold that maximizes the F1 score.

By setting the model's threshold at 0.62, i.e. the point indicated by Youden's J statistic, the FPR is reduced to 0.0%, though at the cost of increasing the FNR to 11.80%. Conversely, setting the threshold at 0.29 (the F1 score point), results in a 0.0% FNR, but FPR rises sharply to 50.0%.

Selecting a threshold point on the ROC curve between these extremes, allows for an optimal trade-off between FPR and FNR, tailored to the specific needs of the clinical application in which the insole system and ML model will be used.

VIII. MODEL EXPLAINABILITY

Explainability is particularly important for models used in medical and clinical settings. A model that can explain its predictions, rather than functioning as a “black box”, helps build trust among all stakeholders. This includes both the end users of the wearable system in which the ML model is deployed and medical professionals interpreting its outputs.

One widely used approach for model explainability is Shapley Additive Explanations (SHAP) [24]. SHAP is model agnostic, meaning it provides explanations independent of

the model's internal workings. It offers both global explanations, which describe how the model behaves overall, and local explanations, which clarify the reasoning behind individual predictions.

SHAP [25] is based on cooperative game theory and aims to fairly attribute the contribution of each input feature (the “players”) to the model's final prediction (the “payoff”). The key steps of SHAP involve: (i) considering all possible subsets of features; (ii) measuring the marginal contribution of a feature s_i by comparing the model's output with and without s_i in each subset; (iii) taking a weighted average of these contributions over all subsets. In our work, we utilise the SHAP Python package [25].

A. Global explanations

Fig. 5 lists the top most important features determined by SHAP. The predominance of right foot sensors in this list, may be influenced by the high collinearity (correlation) between contralateral sensors, as thermal patterns tend to exhibit symmetry across the feet. This strong collinearity could cause the Random Forest model to prioritize one sensor over the other, such as s_8^R over s_8^L .

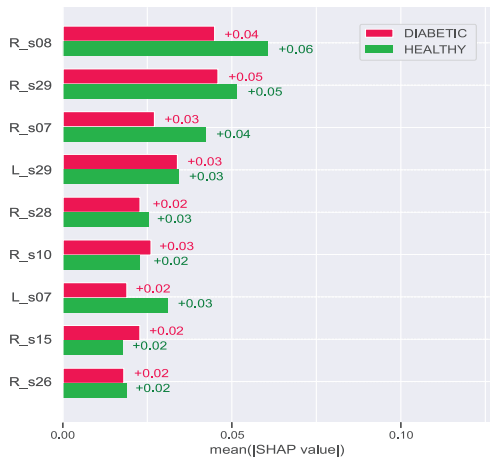


Fig. 5: Feature importance based on mean absolute SHAP values for diabetic (red) and healthy (green) classes.

Also noticeable is the importance of the sensors in the *arch area* of the foot (s_7^R , s_8^R , s_{10}^R , s_7^L). This area is known to be a distinguishing factor for normative (healthy) foot thermal patterns [26].

B. Local explanations

A key property of SHAP values is that they always sum up to the difference between the model's baseline (expected) output and the specific prediction being explained. In other words, the SHAP values of all input features collectively account for the model's deviation from its average prediction.

Fig. 6(b) presents a *waterfall plot* explaining the model's prediction for participant 'A5', a false positive (prediction: diabetic, true value: healthy). The model's initial expected value, $E[f(\mathbf{x})]$, is 0.5, reflecting equal class likelihoods due to the class imbalance correction applied in Section VI.

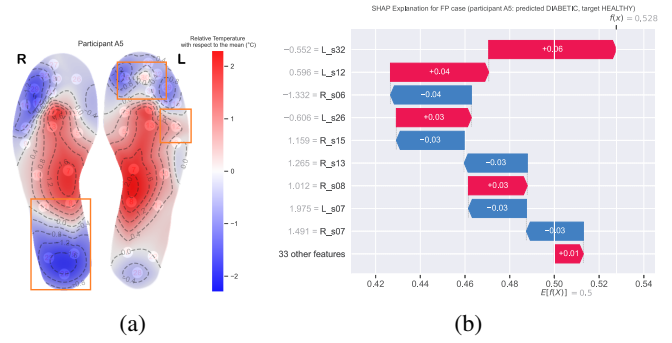


Fig. 6: (a) Temperature data of one of the model's False Positives (FPs), and (b) its SHAP explanation

Among all features, sensor s_{32}^L has the strongest influence, increasing the predicted probability by 6% towards the diabetic class (incorrectly). Similarly, sensors s_{12}^L , s_{26}^L , and s_8^R , also shift the prediction in the same direction, though with smaller contributions. Conversely, the blue-arrowed contributions in Fig. 6(b) pull the prediction toward the correct (healthy) class. However, their combined effect ($\approx 16\%$) is not sufficient to counteract the stronger opposing contributions from the red-arrowed sensors. As a result, the model's final prediction is $f(\mathbf{x}) = 0.53$, surpassing the 0.5 threshold and leading to an incorrect diabetic classification.

Further examination reveals that sensors s_{32}^L , s_{12}^L , and s_{26}^L register warmer-than-expected temperatures – as seen in the regions of the left foot enclosed by yellow boxes in Fig. 6(a), compared to the more normative thermal pattern in Fig. 2(b). Similarly, the right heel appears colder than expected, leading sensor s_8^R to push the prediction towards the incorrect class.

Fig. 7 presents the SHAP explanation for a false negative (FN) case. The strongest incorrect contributions come from sensors s_8^R , s_7^L , and s_7^R , all located in the *medial arch* of the foot. Together, they shift the prediction 18% toward the incorrect class. Although this participant is diabetic, the thermal foot pattern closely resembles one of the normative patterns typically seen in healthy individuals [26]. This aligns with the participant's clinical profile: no PAD, and no history of foot ulceration.

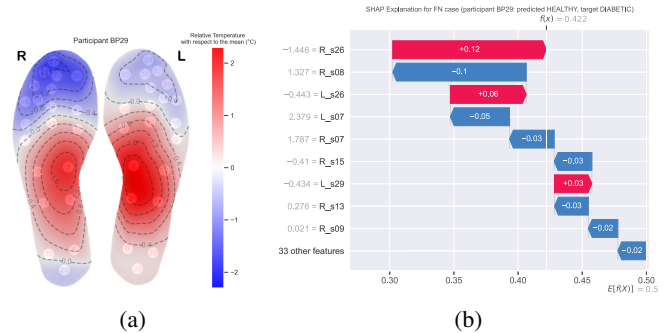


Fig. 7: (a) Temperature data of one of the model's False Negatives (FNs), and (b) its SHAP explanation

The examples above highlight how SCHAP provides clear insights into the model's output, allowing medical

professionals to better understand, validate, and act on its predictions.

IX. ROOT NODE ANALYSIS

To better understand the key features driving the classification decisions, we analyzed the root nodes, i.e. the first splitting points, of the decision trees in the Random Forest model. For each sensor s_i , we identified how many trees used it as the root node, along with the corresponding threshold value and predicted class. Using this information, we constructed “simple classifiers” (decision rules) and assessed their performance. In essence, we reduced each decision tree with s_i as the root node to a *tree stump* (a depth-1 decision tree). We then aggregated their performance results: computing the mean accuracy and confusion matrix. Fig. 8 and 9 illustrate the two top-performing sensor tree stumps, corresponding to sensors s_8^R and s_7^L , both located under the arch of the foot.

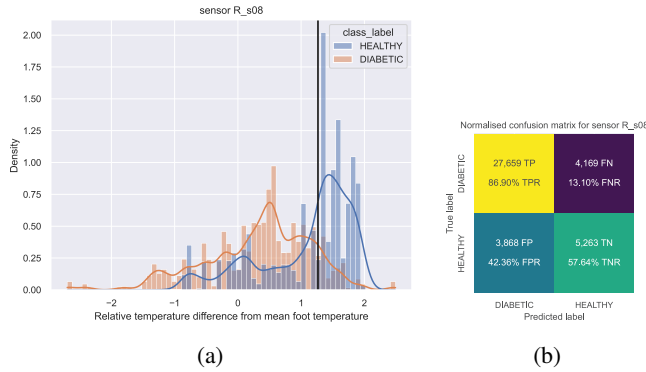


Fig. 8: (a) Class distributions and decision boundary of the decision rule for sensor s_8^R . (b) The confusion matrix of the decision rule.

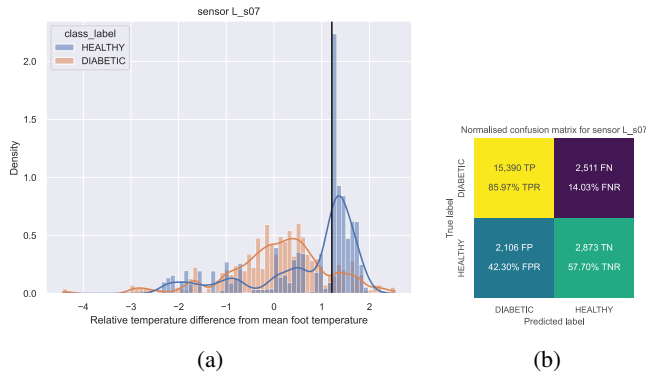


Fig. 9: (a) Class distributions and decision boundary of the decision rule for sensor s_7^L . (b) The confusion matrix of the decision rule.

The tree stumps for sensor s_8^R are characterized by the following decision function, where the threshold value was found to be practically the same across all tree stumps:

$$f(s_8^R) = \begin{cases} \text{DIABETIC}, & \text{if } s_8^R \lesssim 1.266, \\ \text{HEALTHY}, & \text{otherwise.} \end{cases} \quad (2)$$

These simple classifiers achieved a mean accuracy of $80.38\% \pm 0.01\%$. Fig. 8(a) displays the decision boundaries (shown as black vertical lines; appearing practically as a single line due to the nearly identical thresholds of the tree stumps), while Fig. 8(b) shows the aggregated confusion matrix.

The decision function for the tree stumps of sensor s_7^L is defined as:

$$f(s_7^L) = \begin{cases} \text{DIABETIC}, & \text{if } s_7^L \lesssim 1.211, \\ \text{HEALTHY}, & \text{otherwise.} \end{cases} \quad (3)$$

The corresponding classifiers achieve a mean accuracy of $79.82\% \pm 0.02\%$. The decision boundaries and confusion matrix are shown in Fig. 9. The classifiers built using the remaining sensors attained a mean accuracy of 75% or less.

The fact that the simple decision rules (2) and (3) achieve nearly the same accuracy as the full Random Forest model – differing by only 2.82% for s_8^R and 3.88% for s_7^L – suggests that these two features hold substantial predictive power.

From a clinical standpoint, this insight is valuable, as it indicates that a simple threshold-based rule could be nearly as effective as a more complex ML model. Notably, both sensors are positioned in the *medial arch* area of the foot.

Traditionally, temperature-based assessments for diabetic foot complications have relied on asymmetries between the left and right foot [3], [5], analyzing overall temperature distribution patterns [27], or comparing foot temperatures against a standardized ‘healthy foot’ model (template) [12].

In contrast, our approach, which compares localized temperature values to the mean temperature of the same foot, represents a novel and robust alternative that has not been explored to date.

The similarity in threshold values between (2) and (3) suggests a potential diagnostic criterion: if the temperature at the locations of sensors s_8^R and s_7^L in the medial arch area of the foot is less than 1.2°C above the foot’s mean temperature, it may strongly indicate diabetes.

However, before drawing firm conclusions, it is important to consider the moderately low specificity of these rules (57%, cf. Fig. 8(b), Fig. 9(b)). Further validation on a larger, independent dataset is necessary to determine whether this pattern generalizes. Additionally, analyzing the trade-off between sensitivity and specificity would help assess its practical utility. If the single-feature rules (2) and (3) prove robust, they could serve as a simplified screening test or decision aid for clinicians.

X. CONCLUSIONS

In this study, we employed an insole with 21 sensors to capture temperature data from shod feet, focusing on static postures (lying down, sitting, and standing). We utilised relative temperature values with respect to the mean of the foot as the input features for detecting thermal irregularities linked to diabetes-related foot complications – an approach not previously explored.

We trained multiple machine learning classifiers, including Random Forests, Extra Trees, Histogram-based Gradient

Boosting, SVMs, Quadratic Discriminant Analysis, and Logistic Regression. Among these, Random Forest performed best, achieving 83.20% accuracy. This model also demonstrated robustness, with high sensitivity (93.75%) but lower specificity (63.6%). To enhance interpretability, we leveraged SHAP to explain both the model's general behavior and the contributions of individual sensors – a key factor in building trust for clinical use.

Additionally, we extracted simple decision rules by analyzing the root nodes of the Random Forest classifier. The two highest-performing rules, associated with sensors on the medial arch of the foot, achieved accuracies of 80.38% and 79.82%, closely matching the full Random Forest model (83.20%). Unlike traditional temperature-based assessments that focus on left-right asymmetries or overall temperature distribution patterns, these rules leverage the novel use of localized temperature differences relative to the mean of the same foot. This finding can have strong clinical significance, suggesting that a temperature at specific medial arch locations of less than 1.2°C above the foot's mean could serve as a simplified screening test for diabetes.

Future work, will extend this approach to ambulatory activities such as walking, integrating static and dynamic features identified in our related research [18]. Integrating ML models for both static and dynamic postures, can serve as a significant step toward developing an insole-based system for continuous, real-world foot health monitoring.

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